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MILITARY SPECIFICATION

TRAINING AID, AUTOMOTIVE STARTING MOTOR, TYPE MK-7

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of automotive starting motor demonstration training aid, designated Type MK-7.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-527	Enamel Alkyd, Lusterless
TT-P-1757	Primer Coating, Zinc Chromate, Low Moisture Sensitivity

MILITARY

MIL-P-116	Preservation, Methods of
DOD-D-1000	Drawings, Engineering and Associated Lists

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Ogden ALC/MMEDO, Hill AFB, UT 84056 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of

2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein.

PUBLICATIONS

SPECIAL PACKAGING INSTRUCTION (SPI)

000004XA4 - Air Force Fast Pack

(Copies of specifications, standards and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Component parts. The training aid shall consist of a starting motor and a horizontal wood base on which the starting motor shall be mounted.

3.2 Material.

3.2.1 Specifications and standards. Specifications and standards for all materials, parts and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in 3.2.1.1.

3.2.1.1 Standard parts. AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts, such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts list. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.2.1.2 Reclaimed materials. The use of reclaimed material shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

3.2.2 Protective treatment. When materials are used in the construction of the training aid that are subject to deterioration

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when exposed to climatic and environmental conditions likely to concur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3 Design and construction. The training aid shall be designed and constructed so that no parts will work loose in service. The training aid shall be built to withstand the strains, jars, vibrations, and other conditions incidental to shipping, storage, installation, and service.

3.3.1 Maintainability. The training aid shall be constructed so that adjustments and repairs can be easily made by personnel of operating units and overhaul bases.

3.3.2 Starting motor. The starting motor, which shall be furnished by the contractor, shall be the complete unit consisting of the motor, direct-drive manual-shift type drive assembly, housing, and end frames. Portions of the housing and end frames shall be removed to expose as nearly as possible all the internal components of the starting motor. Only those portions of the outer case necessary to support and maintain the components in their respective positions shall be retained. Sectionalization of the starting motor shall not hinder the electrical continuity of the windings of components.

3.3.2.1 Mounting position. The starting motor shall be accurately and firmly mounted on the horizontal base in its normal operating position.

3.3.3 Base. The horizontal wood base shall be constructed of 3/4-inch plywood suitably reinforced to provide a rigid durable structure. The edges of the base shall be processed to prevent the entrance of moisture and shall be trimmed with suitable angle or channel material, as applicable.

3.3.3.1 Size. The base shall extend not more than 2 inches beyond the outside dimensions of the mounted starting motor.

3.3.3.2 Fasteners. Wood screws, bolts, or rivets, where used to attach the angle or channel, shall be flat headed and flush mounted.

3.3.3.3 Rubber pads or feet. Suitable rubber pads or feet shall be attached to the underside of the base to prevent marring of surfaces upon which it may be placed.

3.4 Operation. The training aid shall operate in the same manner as the operational starter motor, except that the training aid shall be manually driven. One revolution of the starting armature or one shift of the drive assembly shall equal one cycle of operation.

3.5 Interchangeability. All parts having the same manufacturer's

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part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000.

3.6 Finishes and protective coatings.

3.6.1 Finish. The training aid shall be finished as specified herein. All enamel used shall conform to TT-E-527 and zinc chromate primer shall conform to TT-P-1757.

3.6.2 Starting motor.

3.6.2.1 Machined surfaces. All machined surfaces exposed by sectionalization as specified herein shall be free from all burrs and marks of machining and shall be finished with one coat of zinc chromate primer and at least two coats of red enamel.

3.6.2.2 Internal surfaces. All internal surfaces of the sectionalized starting motor that are subject to corrosion shall be enameled or plated in a manner so as not to interfere with the functional or instructional value of the machanism. In the event these surfaces are finished with enamel, each internal component part shall be an individual color which shall contrast with all other related components.

3.6.2.3 External surfaces. External surfaces, such as housing and cases, shall be finished with one coat of zinc chromate primer and at least two coats of enamel, Color No. 1645 of FED-STD-595.

3.6.3 Base. All corrosive metal parts on the training aid base shall be finished with one coat of zinc chromate primer. The wood portion of the base shall be sealed, sanded, and then finished with not less than two coats of enamel, Color No. 1610 of FED-STD-595.

3.7 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.8 Workmanship.

3.8.1 General. The training aid, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, riveting, and impregnation of coils; alignment of parts and tightness of assembly screws and bolts; et cetera.

3.8.2 Riveting. Riveting operations shall be carefully performed to insure tightness and satisfactory heading of rivets.

3.8.3 Cleaning. The training aid shall be thoroughly cleaned; and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

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3.9 Government-loaned property. When the contract or purchase order provides, (see 6.2) the Government will loan the following to the contractor for use in fabricating the trainer:

Item No.	Stock No.	Part No.	Item Description	Qty
1	6910-	MIL-T-4775	Training, Aid, Automotive Starting Motor, Type MK-7	1

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. All the tests required for the testing of the trainer are classified as acceptance tests.

4.3 Acceptance tests. Each trainer shall be subjected to the tests as described in 4.4.

4.4 Test Methods.

4.4.1 Examination of product. Each training aid shall be inspected to determine compliance with the requirements of this specification with respect to materials, workmanship, marking, proper operation and performance.

4.4.2 Operation. Each training aid shall be operated through fifty cycles of operation, after which it shall be inspected for the following defects: loose connections, binding, worn parts or undue friction of the moving parts. A training aid found to contain one or more of these defects shall be rejected.

4.5 Inspection of preservation, packaging, packing and marking for shipment and storage. Preparation for delivery shall be inspected in accordance with the requirements of Section 5 or the documents specified herein.

4.6 Rejection and retest. When a trainer fails to meet this specification, it shall not be accepted until the extent and cause of failure have been determined and corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, the acceptance tests shall be repeated.

5. PACKAGING

5.1 Preservation. Each training aid shall be preserved and

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packaged using Method IIC in accordance with MIL-P-116 and Special Packaging Instruction (SPI) 000004XA4.

5.2 Packing. Each training aid packaged in accordance with 5.1 shall be packed in accordance with SPI 000004XA4.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Type MK-7 training aid covered by this specification is intended for use in classrooms to demonstrate the theory and function as well as operation and maintenance of the automotive starting motor.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Method of preservation and interior packing, and if cushioning is required (see 5.2).

6.2.1 Final photographs. The contract or purchase order will specify that three sets of at least 8-by 10-inch glossy, captioned, clear-back, photographic prints of good definition, illustrating the completed training aid and its principal components, shall be delivered to the procuring activity concurrently with or prior to the delivery of the first training aid.

6.2.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian
Air Force - 99

Preparing activity
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